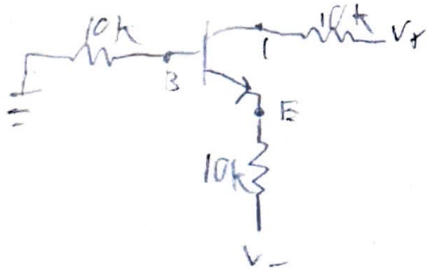

$\beta = 100$ $V_{BEon} = .7V$ Assume active mode

$V_{CEsat} = .2V$

$V_+ = 12V$ $V_- = -12V$ $I_E = 101 I_B$

$0 - 10k(I_B) - .7 - 101 I_B(10k) = -12$

$-1022 I_B = \frac{-11.3}{10k}$

b. $I_B = \frac{-11.3}{10k \cdot (-102)} = \boxed{11.08\,\mu A} = I_B$

$I_E = 101(11.08\,\mu A) = 1.12\,mA$

$I_C = 1.12\,mA\left(\frac{100}{101}\right) = 1.108\,mA$

a. $\frac{0 - V_B}{10k} = 11.08\,\mu A$ $\boxed{V_B = -.111V}$

$\frac{12 - V_C}{10k} = 1.108\,mA$ $\boxed{V_C = .922V}$

$\frac{V_E + 12}{10k} = 1.12\,mA$ $\boxed{V_E = -.811V}$

Our assumption that we are in active mode is correct b/c $V_C > V_B > V_E$ ✓ also $V_+ > V_C > V_B > V_E > V_-$

c. $\boxed{\text{In active mode}}$

d. It does not, I_E is dependent on I_B not I_C or V_C. If I changed the V_- to something like $-15V$, then it would effect I_E.

e. $0 - 10k(I_B) - .7 - 101 I_B(10k) = -12$

$11.1 I_B = \frac{11.3}{10k}$

$I_B = 101.8\,\mu A$

$I_E = 101(101.8\,\mu A) = 10.3\,mA$

$I_C = 10.3\,mA\left(\frac{100}{101}\right) = 10.2\,mA$

$\frac{0 - V_B}{10k} = 101.8\,\mu A$ $\frac{12 - V_C}{10k} = 10.2\,mA$ $\frac{V_E + 12}{1k} = 10.3\,mA$

$V_B = -1.02V$ $V_C = -89.8V$ $V_E = -1.72V$

also not possible

It is in saturation mode

$V_E < V_B$

$V_C < V_B$